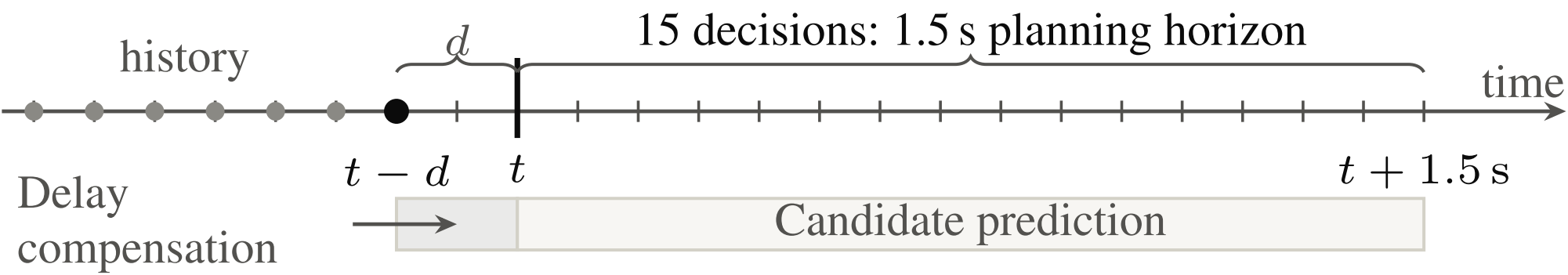
Available information

Delayed: aircraft, load and moving-obstacle states.

Known: goal, pillars, sent setpoints and observation history.

World-model prediction



Use sent setpoints to bridge $t - d \rightarrow t$.

Predict aircraft positions and load position/velocity.

Execution

Planner: 10 Hz; onboard Lee control: 500 Hz on the true state.
Setpoints ramp to their new values over 0.1 s.

Hidden: load mass, cable tensions, controller and motor states.

Oracle U uses true current state; time to scale, delay shown at 200 ms.